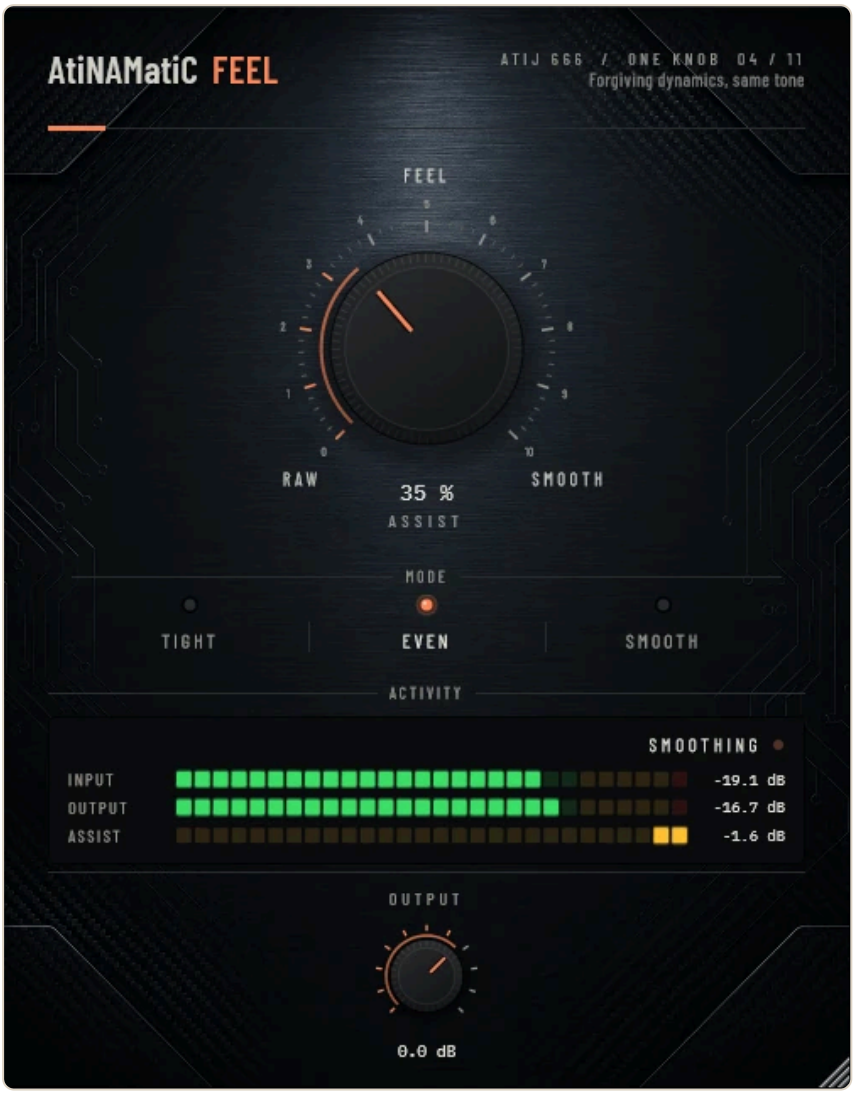
AtiNAMatiC-Feel -- User Manual

Version 1.0.0 · ATIJ STUDIO · Windows · macOS · Linux · VST3 · CLAP · LV2 · AU (macOS) · Free and open source (GPL v3)



What's new in v1.0.0

- First public release, now **free and open source** (GPL v3).
- New look, shared across the whole ONE KNOB Series.
- **Every plug-in format:** VST3, CLAP and LV2 on Windows, macOS and Linux, plus AU on macOS. (The standalone app has been retired: these work best inside your DAW.)
- New **ONE KNOB Series installers** for Windows and macOS: tick the plug-ins and formats you want, skip the rest. Linux gets an install script that asks the same questions.

What's new in v0.2.0

- **Actually tames pick spikes now.** The compressor uses a peak-hold detector with near-instant attack, so spikes come out smaller instead of larger (the old 3.5–18 ms attacks let the first half-millisecond of every attack through at full make-up gain). Still zero latency.
- **No more low-note distortion.** Both detectors used to ripple with the waveform on low notes (up to -22 dB THD on a low E); distortion is now below -60 dB.
- Make-up gain is now calibrated on DI guitar so the level stays within about ±1 dB across the ASSIST sweep; detection is stereo-linked (loudest channel).
- **New interface** — machined knob with a printed 0–10 scale, hardware-style mode switches that follow automation and preset recall, consistent value text ("72 %", "+1.5 dB"), resizable window (75–150 %, remembered per session). Double-click any knob to reset it; hold **Shift** while dragging for fine adjustment.

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1. What it is

AtiNAMatiC-Feel is a **feel enhancer** designed to be placed *before* your amp sim. One large knob shapes how your raw pick attack and dynamics are transformed -- from completely transparent to deeply smoothed -- so you can stay focused on playing while the plug-in quietly makes everything respond better.

Under the hood it is a three-stage processing chain:

- **DC blocker (HPF @ 25 Hz)** -- removes subsonic rumble and DC offset so the compressor envelope is never confused by non-audio content.
- **Transient softener** -- a dual-envelope (1.5 ms fast / 20 ms slow attack, shared release) detector finds the exact spike of each pick attack and applies up to 5 dB of gentle duck, only during the initial milliseconds. Long sustain passes completely unaffected.
- **Feedforward soft-knee compressor (log-domain)** -- a single-sidechain stereo-coherent compressor with an auto-threshold that descends as ASSIST rises. Ratio and knee widen simultaneously, giving a smooth, consistent squeeze rather than sudden gain clamps. Stereo image is preserved because both channels share the same gain envelope from the left/right detection chain.
- **Auto make-up gain** -- static, calibrated on DI guitar (gain reduction at a -4.5 dBFS peak reference), so the output stays within about ±1 dB across the full ASSIST sweep.

Zero latency. No lookahead. No oversampling. Behaves identically to an analogue insert in front of your amp sim.

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2. Signal chain placement

```
Guitar  -->  *** AtiNAMatiC-Feel ***  -->  Amp Sim (NAM / Helix / ML / etc.)  -->  Cab / IR
                                                                                    -->
AtiNAMatiC-Air / Reverb / Master
```

Place Feel **directly before your amp sim**, as the first insert in the chain. This is where it has the most impact -- it shapes the raw pick dynamics before the amp's own non-linearity acts on them.

Not recommended after the cab/IR or in a parallel reverb return -- the transient softener and compressor are tuned for unprocessed guitar signal.

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3. Controls

ASSIST (large centre knob, 0 -- 100%)

The single master macro. Everything is driven by this one control:

ASSIST value	Threshold	Ratio	Knee	Transient softening	Resulting character
0% (RAW)	off	1:1	--	none	fully transparent
35% (default)	~-22 dB	~1.7:1	6 dB	moderate	comfortable, slightly rounded
65% (FEEL)	~-29 dB	~2.3:1	6 dB	substantial	noticeably smoother, forgiving
100% (PILLOW)	~-34 dB	3:1	6 dB	maximum	very compressed, very smooth

The ASSIST pill (bottom-right of the meter panel) shows a label describing the current character: **RAW**, **EASE**, **FEEL**, **SMOOTH**, or **PILLOW**. A teal LED glows brighter as active softening increases.

Tick marks below the knob label the three anchor points: **RAW** (0%), **FEEL** (50%), and **SMOOTH** (100%).

MODE (three-position selector)

Shapes the character of both the transient softener and compressor:

Mode	Comp attack	Comp release	Max ratio	Transient cut	Best for
Tight	near-instant	90 ms	2.5:1	minimal (×0.25)	Articulate playing where you want punch preserved. Palm mutes stay tight and percussive.
Even (default)	near-instant	130 ms	3:1	moderate (×0.6)	Balanced response that works for most styles -- rhythm and lead alike.
Smooth	~0.1 ms	220 ms	4:1	maximum (×1.0)	Heavy softening for laid-back playing, heavily distorted tones, or when you want everything to feel effortless.

OUTPUT (-24 to +12 dB, default 0 dB)

Master output level after all processing. Use it to match the level of the next plug-in in the chain or compensate for any perceived loudness change. The value is displayed inside the knob's text box.

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4. Activity meter and pill

Meter panel (ACTIVITY section)

Three horizontal bars show real-time processing status:

Bar	What it shows
INPUT	Peak level entering the plug-in (post DC blocker)
GR (narrow strip)	Gain reduction in dB, filling right-to-left (up to 12 dB range). Teal colour.
OUTPUT	Peak level after all processing. Colour shifts from grey toward teal as the transient softener becomes active.

The dB scale below the meter runs from -60 dB to 0 dB.

FeelPill (bottom-right of the meter row)

A small LED indicator that shows the current processing character and activity:

Label	ASSIST range	Meaning
RAW	0 -- 5%	No processing
EASE	5 -- 30%	Light touch
FEEL	30 -- 65%	Comfortable zone
SMOOTH	65 -- 90%	Strong smoothing
PILLOW	90 -- 100%	Maximum softening

The teal LED brightens proportionally to how much transient softening is currently active. A fast pick on a bright-sounding string will flash the LED; long sustain will keep it dim even at high ASSIST.

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5. Quick-start settings

Everyday rhythm (feels like a great pick)

- MODE: Even -- ASSIST: 35% -- OUTPUT: 0 dB

Heavy rhythm, lots of palm mutes (tight but forgiving)

- MODE: Tight -- ASSIST: 50% -- OUTPUT: 0 dB

Single-note leads (fluid, consistent sustain)

- MODE: Smooth -- ASSIST: 45% -- OUTPUT: 0 dB

Very aggressive high-gain (tame the harshest attack without killing punch)

- MODE: Tight -- ASSIST: 30% -- OUTPUT: 0 dB

Effortless practice (maximum comfort, nothing hurts)

- MODE: Smooth -- ASSIST: 75% -- OUTPUT: 0 dB

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6. Latency

AtiNAMatiC-Feel introduces **zero latency** at any sample rate.

No lookahead buffer, no oversampling delay, no latency compensation needed. The plug-in reports 0 samples of latency to the host. Your signal is processed in real time, sample by sample.

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7. System requirements

	Windows	macOS	Linux
OS	Windows 10 / 11 (64-bit)	macOS 10.15 Catalina or later	Ubuntu 20.04+ / equivalent
Formats	VST3, CLAP, LV2	VST3, AU, CLAP, LV2	VST3, CLAP, LV2
CPU	x86-64	Apple Silicon (arm64) or Intel x86-64	x86-64
RAM	4 GB minimum	4 GB minimum	4 GB minimum
Sample rates	44.1 -- 192 kHz	44.1 -- 192 kHz	44.1 -- 192 kHz

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8. Installation

Windows (64-bit)

- **Easiest:** run `ONE-KNOB-Series-v1.0.0-Windows-x64-Setup.exe`, tick the plug-ins and formats you want and untick the rest.
- **Just this plug-in:** unzip `AtiNAMatiC-Feel-v1.0.0-Windows-x64.zip` and copy:

File	Copy to
`AtiNAMatiC-Feel.vst3`	`C:\Program Files\Common Files\VST3\`
`AtiNAMatiC-Feel.clap`	`C:\Program Files\Common Files\CLAP\`
`AtiNAMatiC-Feel.lv2`	`C:\Program Files\Common Files\LV2\`

macOS 10.15 or later (Apple Silicon and Intel)

- **Easiest:** open `ONE-KNOB-Series-v1.0.0-macOS-Installer.pkg`, click **Customize** and tick the plug-ins and formats you want.
- **Just this plug-in:** unzip `AtiNAMatiC-Feel-v1.0.0-macOS.zip` and copy:

File	Copy to
`AtiNAMatiC-Feel.vst3`	`/Library/Audio/Plug-Ins/VST3/`
`AtiNAMatiC-Feel.component` (AU)	`/Library/Audio/Plug-Ins/Components/`
`AtiNAMatiC-Feel.clap`	`/Library/Audio/Plug-Ins/CLAP/`
`AtiNAMatiC-Feel.lv2`	`/Library/Audio/Plug-Ins/LV2/`

The plug-ins aren't signed by Apple. If macOS blocks the installer, right-click it and choose **Open**. If you copied files from the zip, clear the quarantine flag in Terminal, e.g. `xattr -cr /Library/Audio/Plug-Ins/VST3/AtiNAMatiC-Feel.vst3`.

Linux (x86_64 and ARM64)

- **Easiest:** extract `ONE-KNOB-Series-v1.0.0-Linux-x86_64.tar.gz` (or `-Linux-ARM64`) and run `./install.sh`. It asks which plug-ins and formats to install. Use `./install.sh --system` to install for all users.
- **Just this plug-in:** unzip `AtiNAMatiC-Feel-v1.0.0-Linux-x86_64.zip` (or `-Linux-ARM64`) and copy the `.vst3` folder to `~/.vst3/`, the `.clap` file to `~/.clap/` and the `.lv2` folder to `~/.lv2/`.

Then rescan plug-ins in your DAW.

9. Controls reference

Control	Range	Default	Description
ASSIST	0 -- 100%	35%	Master macro. Controls compression threshold, ratio, knee, and transient softening simultaneously.
MODE	Tight / Even / Smooth	Even	Selects attack/release time constants and transient-cut intensity.
OUTPUT	-24 to +12 dB	0 dB	Master output level after all processing.

10. Internal signal chain

Each audio channel passes through the following stages, in order, every sample:

```
Input
  └─ DC blocker (SVF high-pass, 25 Hz, Q = 0.5)
       └─ Transient softener
            • Fast envelope follower  (τ ≈ 1.5 ms)
            • Slow envelope follower  (τ ≈ 30 ms)
            • Overshoot ratio  =  (fast − slow) / slow   [clamped 0..1]
            • Transient duck   =  −overshoot × ASSIST × mode_scale × 4 dB  [applied as linear gain]
                 └─ Feedforward soft-knee compressor (log domain, per channel,
                    sidechain from Ch0 for stereo coherence)
                         • Threshold = map(ASSIST, 0→1, −10 dB → mode_floor_dB)
                         • Ratio     = 1 + ASSIST × (mode_ratioMax − 1)
                         • Knee      = mode_kneeDb  (soft, applied at ±knee/2 around threshold)
                         • Envelope  = attack when GR increases, release when GR decreases
                              └─ Auto make-up gain  (static GR at −4.5 dBFS peak reference)
                                   └─ OUTPUT trim
Output
```

The compressor's gain envelope is **shared across all channels** (computed from Ch0, applied everywhere), preserving the stereo image. The DC blocker processes each channel independently.

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11. Typical use cases

Before a NAM amp model

Place Feel as the very first insert before NAM. Set MODE to Even and ASSIST to 30--50%. NAM responds to the dynamics you feed it -- softening the pick transient slightly lets NAM breathe more consistently and removes the harshness that aggressive picking sometimes adds at the amp input stage.

Before a high-gain amp sim (Helix, Neural DSP, etc.)

High-gain amps amplify both your playing and your inconsistencies. Feel at 40--55% ASSIST (Even or Tight mode) smooths out the pick strength variance so the amp responds like you're always in the sweet spot.

Comfort playing / practice

When you want everything to feel effortless -- long sessions, sight-reading, or getting ideas down fast -- set MODE to Smooth and ASSIST to 60--80%. The softener takes the edge off harsh attacks and the compressor keeps dynamics forgiving, so you can focus entirely on the music.

Tracking a lead tone

Leads need consistent sustain without choking the note. Even mode at 45% ASSIST keeps the attack articulate (not squashed) while gently lifting the sustain tail and rounding any sharp pick click.

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12. Tips

- **Start at the default**: MODE Even, ASSIST 35%. This is the calibrated "comfortable" position -- subtle enough that you might not notice it's there, but remove it and you'll miss it.
- **Tight mode for palm mutes**: Tight's short release (80 ms) lets each muted chunk pop distinctly, while still taking the edge off the attack. Great for metal rhythm.
- **Smooth mode for cleans**: On a clean amp, Smooth at 50% ASSIST creates a compressor-pedal-like bloom that suits fingerpicking or chord comping.
- **OUTPUT for gain staging**: If your amp sim or next plug-in is sensitive to input level, use OUTPUT to match unity. Feel's auto make-up covers most of the static GR but you may want a nudge up or down.
- **FeelPill flashing on every pick**: Normal -- it is showing the transient softener firing. If the LED stays bright continuously even between notes, you may have a very hot input signal; lower OUTPUT or the gain into the plug-in.
- **Zero latency means zero offset**: Feel introduces no delay, so no PDC adjustment or manual sample offset is ever needed.

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13. About

AtiNAMatiC-Feel is part of the AtiNAMatiC series by **Atij 666 Studio** -- a collection of guitar-focused plug-ins built around a single principle: one knob that does the right thing.

Built with: - JUCE -- Raw Material Software Ltd. - VST3 SDK -- Steinberg Media Technologies GmbH - clap-juce-extensions -- Free Audio - Neural Amp Modeler (NAM) -- Steven Atkinson & contributors

Thank you to everyone who makes open-source audio tools possible.

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